Indirect reciprocity with inertial reputation updating

Introduction, Storyline 3: from information to judgement

Cooperation among unrelated individuals is sustained in part by reputation[1,2]. By helping others, individuals improve their standing and become more likely to receive help from third parties, a mechanism known as indirect reciprocity[3–6]. Theory has identified social norms that make reputation-based cooperation evolutionarily stable[7,8], most notably the leading-eight norms, which sustain cooperation when reputations are public, even in the presence of occasional errors[8,9]. These results rest on agreement about who is good and who is bad. When reputations are private and assessments are noisy, individuals hold different views of the same person[10,11], and most leading-eight norms no longer maintain stable cooperation[12–15]. A central problem for indirect reciprocity is therefore how reputations can remain both accurate and shared under private assessment. They must distinguish those who cooperate from those who do not, and different observers must largely agree about who is who[16,17].

The difficulty is sharpest for the norms that sustain cooperation under public information. Under these norms, a discriminator who refuses to help an individual in bad standing keeps its own good standing, because the refusal counts as justified punishment[7,8,18]. This exemption works only if the observer shares the donor's view of the recipient, which under private assessment is often not the case. An observer who still regards the recipient as good sees an unjustified refusal and condemns the discriminator. Disagreement about one individual thereby becomes disagreement about those who act towards it, and it spreads through the population[12,19,20]. Discriminators thus face a punishment dilemma. Withholding help from defectors protects cooperation against exploitation, yet each act of punishment risks their own reputation in the eyes of observers who disagree[18,21,22]. The dilemma can be eased in two ways. Observers can be brought to agree about each other's standing, so that justified punishment is recognized as such, or reputations can be made robust enough that a single disputed judgement does not overturn them. In both cases, genuine defectors must still lose their standing.

Existing remedies pursue both routes mainly by acting on the information that observers hold. They make it shared, through public institutions, gossip or the synchronization of private opinions[17,23–25], or richer, through scores, records or the aggregation of several observations[21,26,27]. Better information reduces erroneous and disputed judgements but does not eliminate them, because observation and assessment remain noisy[12,15]. What remains is a problem of judgement, namely how a standing verdict about another person should be revised as new evidence arrives. Any revision rule faces a trade-off between condemning cooperators on disputed evidence and excusing defectors whose misconduct is not yet conclusive. Human institutions often manage this trade-off through procedure. They escalate sanctions only with repeated violations[28,29] and restore an offender's standing only after a period of good conduct[30], and impressions of other people are likewise revised asymmetrically[31–34]. Models of indirect reciprocity, by contrast, mostly revise reputations through a single symmetric rule, whether the latest assessment, a score or a fixed window of observations[8,12,21,26]. How the rule by which private judgements are revised shapes the punishment dilemma, and how such a rule should be designed, remains open.

Here, we develop a model of indirect reciprocity in which observers revise their private judgements with inertia (Fig. 1). An established reputation changes only after a run of contrary assessments, and the run required to lose standing can differ from the run required to regain it. Negative assessments move an individual in good standing through warning states, and positive assessments move an individual in bad standing through recovery states. Each intermediate state can already carry partial consequences,

because it is interpreted as good or bad with an adjustable probability. Observers share no information and update on one observation at a time, so any agreement among them must emerge from how they revise their judgements.

We find that the direction of the revision rule, rather than the amount of evidence it requires, decides whether private observers agree with one another and still distinguish cooperators from defectors. The beneficial direction depends on the norm, mirroring the kind of misjudgement that private assessment produces under each norm. Compact rules, selected solely by these reputational criteria, make cooperation stable and easy to reach under four classical norms, whereas the same amount of inertia placed on the wrong path can leave no cooperation at all. The rule that best protects cooperators from disputed evidence, however, also shelters opportunists who defect only occasionally, a weakness that the aggregation and score-based rules we test share. Allowing warning states to carry partial consequences restores deterrence while cooperation remains high. Sustaining cooperation through private reputations is therefore not only a matter of what individuals know about each other, but also of how they revise what they believe.

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